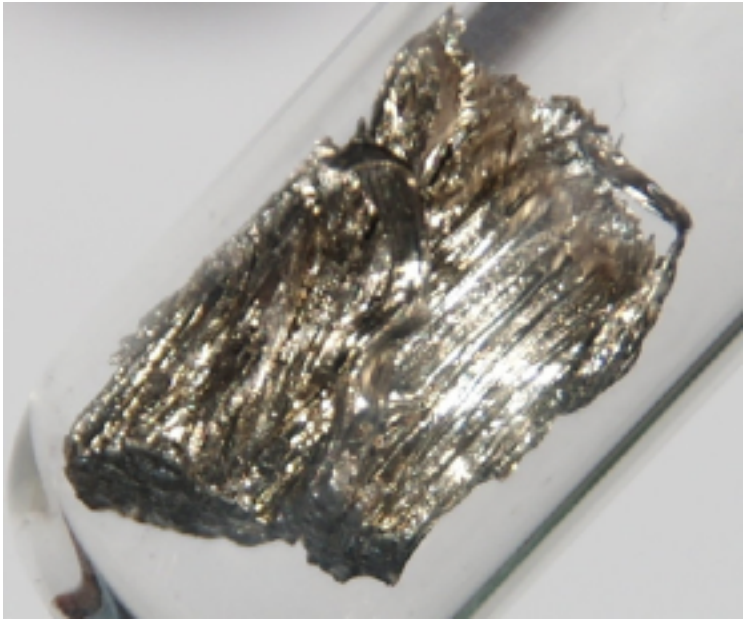




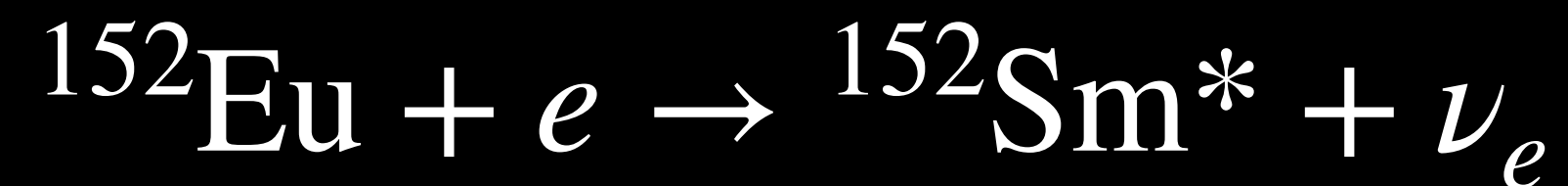






What are these nuclei *A* and *B*?

- ★ $A = {}^{152m}\text{Eu}$ with $J^\pi = 0^-$
- ★ The nucleus ${}^{152m}\text{Eu}$ captures a *K*-shell electron and transforms into an excited samarium nucleus ${}^{152}\text{Sm}^*$ with $J^\pi = 1^-$



- The excited nucleus ${}^{152}\text{Sm}^*$ decays after about 0.07 ps to the ground state, emitting a γ ray before the recoil-direction information is lost



${}^{152}\text{Eu}$



${}^{152}\text{Sm}^*$

The result

$$h(\nu) = -1 \pm 0.3$$



Moris Goldhaber (1937)
Source: [wikipedia](https://en.wikipedia.org/wiki/Moris_Goldhaber)